Akash Gogate

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Education

University of Wisconsin-Madison **Graduating May 2028**

Bachelor of Science in Computer Science & Biology — Dean's List Fall 2024, Spring 2025, Fall 2025 — 3.8 GPA

Relevant Coursework: Advanced Algorithms, Data Structures, Artificial Intelligence, Deep Learning, Machine Organization & Programming, Probability & Statistics, Computer Science Capstone

Technical Skills

Languages: Java, Python, Rust, TypeScript, C++, R, SQL, Linux, Bash
Systems & Tools: PyTorch, TensorFlow, Scikit-learn, Kafka, Docker, Kubernetes, MongoDB, DuckDB, FastAPI, REST APIs, GitHub Actions, Neo4j, OpenCV, Weights & Biases

Experience

Leidos **May 2025 – Present**

Software Engineer Intern

- Built **Neo4j** knowledge graph with LLM plugin at **300–500** nodes for **30-engineer** team onboarding.
- Built **Kafka+MongoDB** pipeline at thousands of events/sec with dead-letter queue recovery.
- Automated regression testing via **Docker** & **GitHub Actions** across 5 system components.
- Eliminated **200+** satellite scheduling conflicts via **Python** memoization & DP across **1,000+** active satellites.
- Built **DuckDB** analytics with **LLM** query interface at sub-second latency across fleet readiness datasets.
- Implemented **Rust** microservice for satellite scheduling validation cutting manual review time.

The Kendziorski Lab at UW-Madison **Sep 2025 – Present**

Machine Learning Research Intern

- Reduced scRNA-seq pipeline runtime **80→27 min** via **PyTorch** DataLoader optimization across **4** architectures.
- Cut manual literature review ~**60%** via NLP entity extraction across **500+** papers.
- Validated convergence across **4** architectures via **W&B** for Nature Bioinformatics co-authorship.

Inspirit AI Research Internship **Sep 2022 – Mar 2023**

Machine Learning Research Intern

- Achieved **95%** skin cancer accuracy via **Decision Tree** on **70K** dermoscopic images using **scikit-learn**.
- Trained **Random Forest** on miRNA sequences at **95%** accuracy ($p<0.05$) in **R** via stratified k-fold CV.

Projects

Self-Improving LLM Agent | Github Link **Jan 2026 – May 2026**

- Architected failure-driven self-improvement loop outperforming ReAct **15pp** on 5,000+ tasks via LLM-as-judge harness.
- Built LLM serving layer across **Claude**, **Grok**, **Llama** & **Qwen** with unified adapter interfaces.

Lotus Health | Github Link **March 2026**

- Built ICD-10 risk engine on 1,080-node graph over **45M** records via **Groq Llama** RAG.
- Designed hybrid **Groq Llama** / Python ICD-10 scoring for reproducible clinical outputs.
- Integrated **Deepgram** streaming transcription for real-time voice-to-ICD-10 risk scoring over **45M** records.

LLM-Based SAT Practice Test Generator | Github Link **Jun 2024**

- Built full-stack SAT prep app in **TypeScript** & **FastAPI** with **GPT-4o** adaptive question generation & scoring.
- Designed REST API in **FastAPI** serving **75** cached SAT questions across **3** sections by user performance history.

Ecological Conservation Game | Github Link **Jan 2024**

- Engineered **C++** simulation in **Raylib** sustaining **60+ FPS** across **10,000+** agents via O(N) spatial hashing.
- Engineered **4**-species emergent ecosystem with trait inheritance & fitness selection across **10,000+** concurrent agents.

Leadership

Princeton Racket Club **May 2024 – Aug 2024**

Tennis Coach & Tournament Director

- Directed **8** regional tournaments at **380+** match entries with 100% on-time rate as dual director & head coach.
- Coached **8–10** athletes per class in **3** cohorts, advancing **3 juniors 150+** ranking pts & **5 adults 0.5 NTRP**.

TEL: Spatial Intelligence Ideathon **July 2026**

- Won **2nd place** among **50+** teams via **ZaiNAr** sub-meter **AR** for wheelchair-accessible airport wayfinding.
- Pitched dual-product airport platform to **5** judges via **ZaiNAr** precision tracking for airline ops.